

Abhimanyu Singh

Programmer Analyst | AI/ML Developer | Data Scientist

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Profile

Generative AI Developer with 4+ years of experience designing, developing, and deploying end-to-end GenAI and Agentic AI pipelines combined with Natural Language Processing (NLP) techniques and Computer Vision systems. Proven track record of, improving automation workflows, and delivering production-ready Agentic AI solutions for enterprise clients including Cox Communications, Pfizer, and Xerox. Holds a B.Tech in Computer Science.

Professional Experience

Programmer Analyst , <i>Cognizant Technology Solutions</i>	02/2025 – Present Noida, Uttar Pradesh
- Automated helpdesk operations and ticket resolution workflows using Generative AI. - Designed intelligent solutions reducing manual effort by 80% with an accuracy of over 90%. - Integrated Databases, Knowledge Graphs, and Langchain/LangGraph to build conversational AI assistants.	
Software Engineer , <i>HCLTech</i>	07/2022 – 02/2025 Noida, India
- Developed AI-based document classification system using OCR and Transformer based classification models. - Experimented with LLMs for summarization, text generation, and Retrieval Augmented Generation (RAG). - Gained hands-on experience working with open-source vision-based LLMs.	

Education

Bachelor of Technology - Computer Science , <i>Rajasthan Technical University</i>	07/2018 – 07/2022 Jaipur, Rajasthan, India
Percentage: 84%	
Higher Secondary , <i>Brij Bal Sr Secondary School</i>	03/2017 – 07/2018 Jaipur, Rajasthan, India
Percentage: 74%	

Skills

Programming & Frameworks

Python, NumPy, pandas, Scikit-learn, Tensorflow, PyTorch, Transformers, OpenCV, Flask, FastAPI, Streamlit

AI/ML & Deep Learning

Supervised & Unsupervised Machine Learning, Computer Vision, CNN, RNN, ResNet, OCR, CUDA

NLP & Generative AI

LLMs, BERT, ROBERTa, NLTK, Tokenization, TF-IDF, Embeddings, Prompt Engineering, LangChain, LangGraph, LlamaIndex, RAG, Vector DBs, Semantic Search, Model Context Protocol, (MCP), LLMOps, Agentic AI, PageIndex, VectorLess RAG

Cloud & DevOps

AWS, Azure, Docker, Git, CI/CD, Kubernetes

Projects

Root Cause Analysis Agent, *Cox Communications*

- Designed an RCA Agent to automate application-level troubleshooting using log analysis.
- A LangGraph-based agent that generates detailed RCA with recommended actions and actionable insights by just using the job log file and the relevant python script file.
- Reduced LLM costs by upto 50% using advanced prompt caching techniques and smart truncation of the log file contents..
- Reduced manual effort by 80% and improved ticket resolution SLA compliance.
- Tech Stack: Langgraph, ChromaDB, Python, FastAPI, AWS Bedrock, Claude Opus 4.6

Hybrid Retrieval-Augmented Generation (RAG) knowledge Engine

- Built a custom Hybrid RAG pipeline for enterprise knowledge management using Langchain & ChromaDB.
- Enhanced accuracy of LLM response through document chunking, embeddings, and hybrid retriever.
- Supported multi-model inputs (text, PDFs scanned documents) for scalable enterprise deployments.
- Tech Stack: Langchain, Chromadb, Azure OpenAI, FastAPI

Intelligent Document Processing (Document AI), *Xerox*

- Developed an end-to-end Document Intelligence system combining OCR+NLP+RAG pipelines for structured data extraction.
- Applied PaddleOCR & Tesseract for text recognition, Spacy/NLTK for NER, and Langchain + FAISS for contextual search.

Integrated Responsible AI guardrails:

- PII masking & redaction for sensitive data.
- Input validation & response moderation to handle toxicity, bias, and prompt injection attacks.
- Hallucination control based on document context similarity score.
- Achieved 95% entity recognition accuracy and automated data validation workflows, reducing manual effort by 70%.
- Tech Stack: Python, PaddleOCR, Spacy, Langchain, Guardrails, FAISS, SQL, HuggingFace Transformer models.

Certificates

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| • Introduction to Large Language Models | • Developing AI Applications with Python and Flask | • Complete Agentic AI Bootcamp with Langgraph and Langchain |
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